

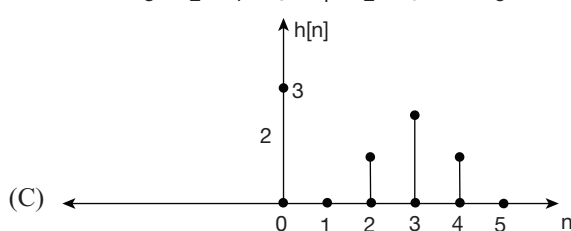
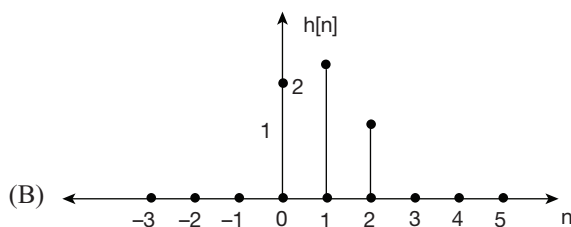
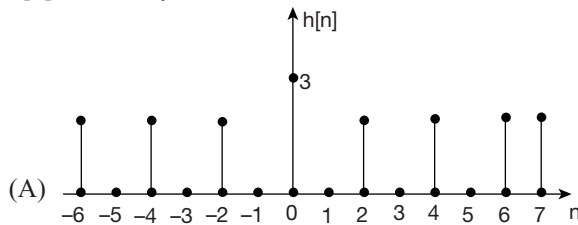
SIGNALS AND SYSTEMS TEST 4

Number of Questions: 25

Time: 60 min.

Directions for questions 1 to 25: Select the correct alternative from the given choices.

- A system is formed by cascading two impulse responses $h_1(t)$ and $h_2(t)$ given as
 $h_1(t) = e^{-3t} u(t)$
 $h_2(t) = 2e^{-5t} u(t)$
 The overall system impulse response $h(t)$ and overall system is _____
 (A) $(e^{-3t} + e^{-5t}) u(t)$ and BIBO stable
 (B) $(e^{-3t} - 2e^{-5t}) u(t)$ and BIBO stable
 (C) $[e^{-3t} - e^{-5t}] u(t)$ and BIBO stable
 (D) $2[e^{-3t} - e^{-5t}] u(t)$ and BIBO unstable
- Two signals
 $x_1[n] = \{1, 2, 5\}$
 $x_2[n] = \{1, 1, 1, 1, 1, 1\}$
 Convolution $y[n] = x_1[n] \otimes x_2[n]$ is
 (A) $\{1, 3, 8, 8, 8, 7, 5\}$ (B) $\{1, 2, 5\}$
 (C) $\{1, 3, 8, 8, 8, 7, 5\}$ (D) None of the above
- The convolution $(x[n] + 5\delta[n-5]) \otimes \delta[n-5]$ can be simplified to –
 (A) $x[n-5] + 5\delta[n-5]$ (B) $x[n] + 5\delta[n-5]$
 (C) $x[n-5] + 5\delta[n-10]$ (D) $x[n-5]$
- Which of the following system with impulse response $h[n]$ is memory less?



(D) None of the above

- A system with impulse response $h[n] = a^n b^n u[n]$ is stable if

- $|a| > 1$
- $|b| < 1$
- $|a| |b| < 1$
- For all values of a and b system is stable

- A discrete sequence is given as
 $x[n] = \{-3, -3, -2, 2\}$

↑
 the auto-correlation $r_{xx}[n]$ is –

- $\{-6, 0, 11, 26, 11, 0, -6\}$
- $\{-6, 0, 11, 26, 11, 0, -6\}$
- $\{-6, -1, 0, 22, 11, 0, -6\}$
- $\{-3, -3, -2, 2, 2, -2, -3, -3\}$

- The impulse response of a system is given by –

$$h[n] = 5 \left(\frac{-1}{3} \right)^n u[n] - 3 \left(\frac{-1}{4} \right)^n u[n]$$

For this system two statements are

- System is causal and stable
- Inverse system is causal and stable
- 1 is true (B) 2 is true
- Both are true (D) Both are false

- Number of calculation reduced in radix – 2 FFT in comparison to N -point DFT is –

- $N \log_2 N$ complex addition and $\frac{N}{2} \log_2 N$
- $N(N-1)$ complex addition and N^2 complex multiplication.
- N complex addition and $\frac{N}{2}$ complex multiplication.
- None of these

- In decimation in time algorithm for Radix – 2 FFT, the N -point DFT can be realized from

- One number of $\frac{N}{2}$ point DFT.
- Two number of $\frac{N}{2}$ point DFTs.
- One number of N -point DFT.
- Two number of N -point DFTs.

- Circular convolution of two sequence

$$x_1[n] = \{2, 3, 1, 1\}$$

$$x_2[n] = \{1, 3, 1, 2\}$$

↑
 is

- | | | | |
|-----|------------------|-----|------------------|
| (A) | [14, 13, 12, 11] | (B) | [11, 12, 13, 14] |
| | ↑ | | ↑ |
| (C) | [13, 14, 11, 12] | (D) | [12, 14, 11, 12] |
| | ↑ | | ↑ |

11. An LTI discrete system having input $x[n]$ and output $y[n]$ is described by following difference equation
 $y[n] = x[n] + x[n - 3] + x[n - 5]$
 The transfer function $H(e^{j\Omega})$ of the system is ____

- (A) $e^{-4j\Omega} (1 + e^{j\Omega} + e^{-j\Omega})$
 (B) $e^{-4j\Omega} \left(1 + \frac{\cos \Omega}{2}\right)$
 (C) $e^{-4j\Omega} (2 + \cos \Omega)$
 (D) $1 + 2 \cos \Omega e^{-4j\Omega}$

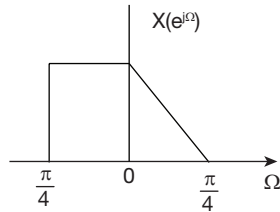
12. The 4-point discrete Fourier Transform of a discrete sequence $x[n]$ is given as

$$X_{DFT}[k] = \{4, -j, 0, j\}$$

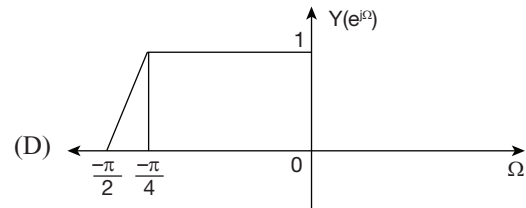
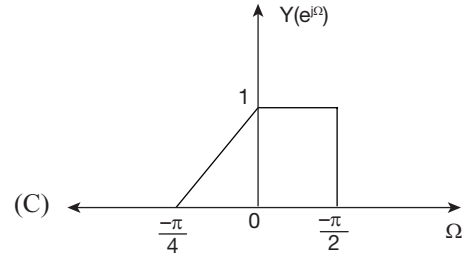
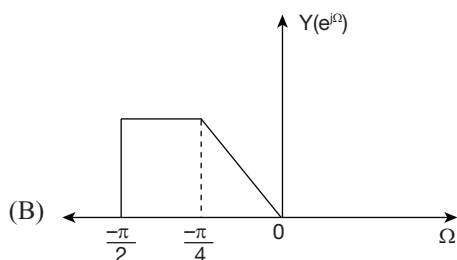
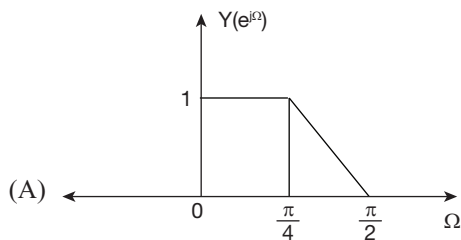
The sequence is _____

- (A) $\{1.5, 1.5, 1, 0\}$
 (B) $\{1, 1.5, 1, 0.5\}$
 (C) $\{1, 0.5, 1, 0.5\}$
 (D) $\{1, 1.5, 1.5, 0\}$

13. The Zero-phase DTFT of a sequence $x[n]$ is shown in figure below –



Which one of the following is the DTFT of the sequence $y[n] = x[n] e^{-jn\frac{\pi}{4}}$?



14. Match the following according to their appropriate Fourier representation of the signals.

List-I	List-II
(a) $x[n] = \left(\frac{1}{3}\right)^n u[n]$	(i) Fourier Series
(b) $x(t) = 1 + \cos 3\pi t + \sin 5\pi t$	(ii) DTFT
(c) $x(t) = e^{-2t} \cos 4\pi t u(t)$	(iii) FT
(d) $x[n] = \sum_{m=-\infty}^{+\infty} \delta[n-20m] - 2\delta[n-3-20m]$	(iv) DTFS

- (A) $a - (\text{ii}), b - (\text{i}), (c) - (\text{iii}), d - (\text{iv})$
 (B) $a - (\text{iv}), b - (\text{i}), (c) - (\text{ii}), d - (\text{iii})$
 (C) $a - (\text{i}), b - (\text{ii}), (c) - (\text{iii}), d - (\text{iv})$
 (D) $a - (\text{ii}), b - (\text{iii}), (c) - (\text{iv}), d - (\text{i})$

15. A continuous-time ideal band pass filter whose frequency response is

$$H(j\omega) = \begin{cases} 1, & 2\omega_c \leq |\omega| \leq 4\omega_c \\ 0, & \text{Otherwise} \end{cases}$$

If $h(t)$ is impulse response of this filter then the function $g(t)$ is _____

$$\text{If } h(t) = \left(\frac{\sin \omega_c t}{\pi t} \right) g(t)$$

- (A) $2 \cos 2\omega_c t$
- (B) $2 \cos 3\omega_c t$
- (C) $2 \cos \omega_c t$
- (D) $\frac{\cos 3\omega_c t}{2}$

16. If the group delay of a discrete time *LTI* system with frequency response $H(e^{j\omega})$ and real impulse response is given as

$$\tau\left(\frac{\pi}{2}\right)=2 \& \left|H\left(e^{\frac{j \pi}{2}}\right)=2\right|$$
. Then the input of the

system, if output is $2 \cos\left(\frac{\pi}{2}n - \frac{3\pi}{4}\right)$ is _____

- (A) $\cos\left(\frac{n\pi}{2} + \frac{\pi}{4}\right)$ (B) $\cos\left(\frac{n\pi}{4} + \frac{7\pi}{4}\right)$
 (C) $\cos\left(\frac{n\pi}{2} - \frac{7\pi}{2}\right)$ (D) $\cos\left(\frac{n\pi}{2} - \frac{\pi}{4}\right)$

17. A discrete time LTI system with impulse response $h[n]$ is given by –

$$h[n] = \beta^n u[n]$$

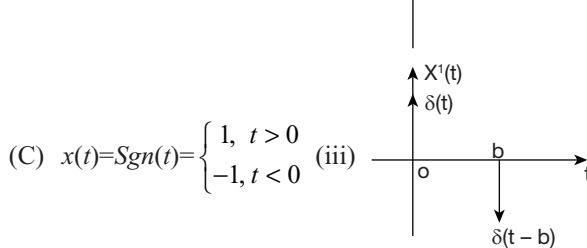
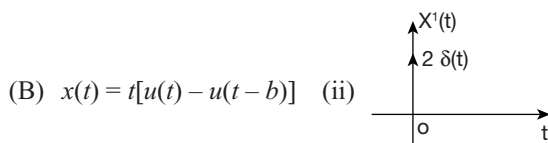
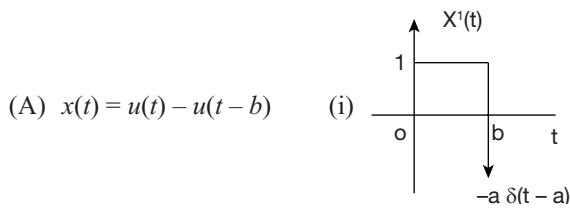
This system is

- (A) Causal and BIBO stable if $\beta > 1$
 (B) Non-causal and BIBO unstable if $\beta < 1$
 (C) Causal and BIBO unstable if $\beta \geq 1$
 (D) Non causal and BIBO stable if $\beta < 1$

18. Match the List-II with the first derivatives of following signals –

List - I

List- II



- (A) (a) – (iii), b – (i), (c) – (ii)
 (B) (a) – (ii), b – (ii), (c) – (i)
 (C) (a) – (iii), b – (ii), (c) – (i)
 (D) (a) – (i), b – (ii), (c) – (iii)

19. (i) $y[n] = x^2[n]$
 (ii) $y[n] = \log\{|x[n]|\}$
 (iii) $y[n] = \sin\{x[n]\}$
 (iv) $y[n] = x[n] + x[n-1]$

Invertible DT system is –

- (A) (i) and (ii) only (B) (ii) and (iv) only
 (C) (iii) only (D) (iv) only

20. A discrete time system with output $y[n]$ and input $x[n]$ is given. Causal System is _____

- (A) $y[n] = 3^{n+2} x[n]$ (B) $y[n] = 3^n x[n+1]$
 (C) $y[n] = 3^n x[n]$ (D) Both A and C

21. A Discrete time system is described by –

$$y[n] = \begin{cases} 5x[2n] & x < 0 \\ 0 & x \geq 0 \end{cases}$$

Consider the following properties

P_1 : System is Causal

P_2 : System is Linear

P_3 : System is invertible

The above properties processed by the system is _____

- (A) P_1, P_2 (B) P_2, P_3
 (C) P_1 only (D) None of the above

22. A linear time-invariant system has the following system function

$$H(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} + \frac{4}{1 - 2z^{-1}}$$

- (1) The system is stable if ROC: $|z| > \frac{1}{3}$
 (2) The system is causal if ROC: $|z| > \frac{1}{3}$
 (3) The system is stable if ROC: $\frac{1}{3} < |z| < 2$
 (4) The system is causal if ROC: $|z| > 2$

Which of the statement is/are correct?

- (A) 1 and 2 (B) 1 and 4
 (C) 2 and 3 (D) 3 and 4

23. Which of the following causal LTI system with system function $H(s)$ is stable?

- (A) $\frac{(s-3)}{(s+2)(s-1)}$ (B) $\frac{(s-3)}{s^2(s+3)(s-1)}$
 (C) $\frac{(s-3)}{s(s+3)(s+2)}$ (D) None of these

24. The 4-point DFT of causal three sample sequence is

$$\text{given by } x[n] = \begin{cases} \frac{1}{3}, & 0 \leq n \leq 2 \\ 0, & \text{else} \end{cases}$$

- (A) $\left\{\frac{j}{3}, 1, \frac{-j}{3}, \frac{1}{3}\right\}$ (B) $\left\{1, \frac{-j}{3}, \frac{1}{3}, \frac{j}{3}\right\}$
 (C) $\left\{1, \frac{-j}{3}, -1, \frac{j}{3}\right\}$ (D) $\left\{1, \frac{+j}{3}, +1, \frac{-j}{3}\right\}$

25. If $X[k]$ is the 4-points DFT of a sequence $x[n]$, then the DFT of the sequence $y[n] = x[n] \cos 2\pi n$ is –

- (A) $\frac{1}{2} [X(k+2) - X(k-2)]$
 (B) $\frac{1}{2} [X(k-1) + X(k+1)]$
 (C) $\frac{1}{2} [X(k-1) - X(k+1)]$
 (D) $\frac{1}{2} [X(k-2) + X(k+2)]$

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. C | 3. C | 4. D | 5. C | 6. B | 7. C | 8. A | 9. B | 10. D |
| 11. D | 12. B | 13. B | 14. A | 15. B | 16. A | 17. C | 18. A | 19. D | 20. D |
| 21. D | 22. D | 23. C | 24. B | 25. B | | | | | |

HINTS AND EXPLANATIONS

Solutions for questions 1 to 25:

$$1. H_1(\omega) = \frac{1}{j\omega + 3}$$

$$H_2(\omega) = \frac{2}{j\omega + 5}$$

$$H(\omega) = H_1(\omega) H_2(\omega) = \frac{2}{(j\omega + 3)(j\omega + 5)}$$

$$H(\omega) = \left[\frac{1}{j\omega + 3} - \frac{1}{j\omega + 5} \right]$$

$$h(t) = [e^{-3t} - e^{-5t}] u(t)$$

For BIBO stability

$$\int_{-\infty}^{+\infty} h(t) dt = \int_0^{\infty} (e^{-3\tau} - e^{-5\tau}) d\tau$$

$$= \left[\int_0^{\infty} e^{-3\tau} d\tau - \int_0^{\infty} e^{-5\tau} d\tau \right]$$

$$= \left[\frac{1}{3} - \frac{1}{5} \right] = \frac{2}{15} < \infty$$

Thus the system is BIBO stable.

Choice (C)

2.

$X_{2[n]}$	1	1	1	1	1	1
$X_{1[n]}$	1	1	1	1	1	1
2	2	2	2	2	2	2
5	5	5	5	5	5	5

$$y[n] = \{1, (2+1), (5+2+1), (5+2+1), (5+2+1), (5+2+1), (5+2), 5\}$$

$$= \{1, 3, 8, 8, 8, 8, 7, 5\}$$

Choice (C)

3. From the distribution property of convolution we can write –

$$(x[n] + 5\delta[n-5]) \otimes \delta[n-5]$$

$$= x[n] \otimes \delta[n-5] + 5\delta[n-5] \otimes \delta[n-5]$$

$$= x[n-5] + 5\delta[n-10]$$

Choice (C)

4. For a memory less system

$$h[n] = 0 \text{ for } n \neq 0$$

But for all impulse responses given here

$$h[n] \neq 0 \text{ for } n \neq 0$$

Choice (D)

5. For stable system impulse response should be

$$\sum_{n=-\infty}^{+\infty} |h[n]| < \infty$$

Since $u[n] = 1$

$$\text{So } \sum_{n=0}^{\infty} |h[n]| < \infty$$

$$\sum_{n=0}^{\infty} |a|^n |b|^n = 1 + |a||b| + |a|^2 |b|^2 \dots$$

$$= \frac{1}{1 - |a||b|}$$

The R.H.S converges if $1 - |a||b| > 0$

$$|a||b| < 1$$

Choice (C)

6. Auto-correlation is given by –

$$r_{xx}[n] = x[n] \otimes x[-n]$$

$$x[n] = \{-3, -3, -2, 2\}$$

$$\uparrow$$

$$x[-n] = \{2, -2, -3, -3\}$$

$$\uparrow$$

the convolution table is shown below –

$\rightarrow x(n)$	-3	-3	-2	2
$\downarrow x(-n)$				
2	-6	-6	-4	4
-2	6	6	4	-4
-3	9	9	6	-6
-3	9	9	6	-6

$$r_{xx}[n] = \{-6, 0, 11, 26, 11, 0, -6\}$$

Choice (B)

$$7. H(z) = \frac{5}{1 + \frac{1}{3}Z^{-1}} - \frac{3}{1 + \frac{1}{4}Z^{-1}}$$

$$= \frac{5 + \frac{5}{4}Z^{-1} - 3 - Z^{-1}}{\left(1 + \frac{1}{3}Z^{-1}\right)\left(1 + \frac{1}{4}Z^{-1}\right)} = \frac{2 - \frac{1}{4}Z^{-1}}{\left(1 + \frac{1}{3}Z^{-1}\right)\left(1 + \frac{1}{4}Z^{-1}\right)}$$

Both the poles of this system are inside $|Z| = 1$.

So the system is stable and causal.

For the inverse system also all the poles are inside $|Z| = 1$. So inverse system is also stable and causal.

Choice (C)

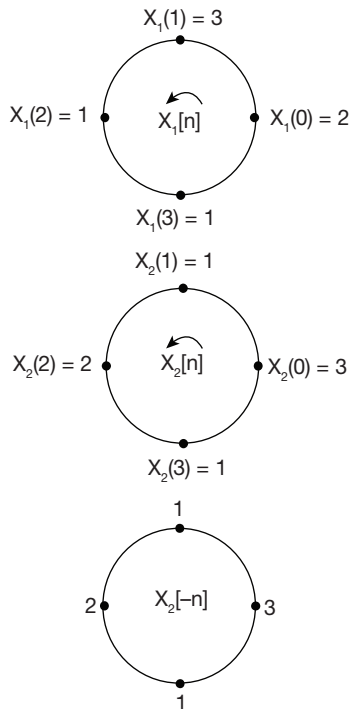
8. In Radix – 2 FFT with m stages and each stage having $\frac{N}{2}$ butterflies and each butterfly involve one complex multiplication and two complex additions.

So the number of computations in each stage are $\frac{N}{2}$ complex multiplication and N complex addition. So $\frac{N}{2} \times m$ or $\frac{N}{2} \log_2 N$ complex multiplication and $N \log_2 N$ complex additions. Choice (A)

9. N -point DFT in DIT can be realized from two numbers of $\frac{N}{2}$ point DFTs.

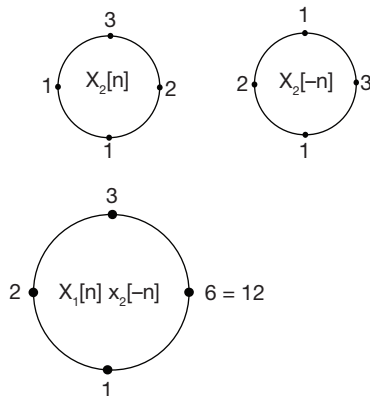
The $\frac{N}{2}$ point DFT can be realized from two numbers of $\frac{N}{4}$ point DFTs and so on. Choice (B)

10.

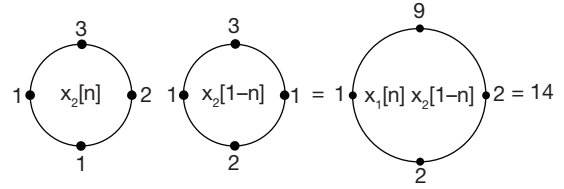


Now when

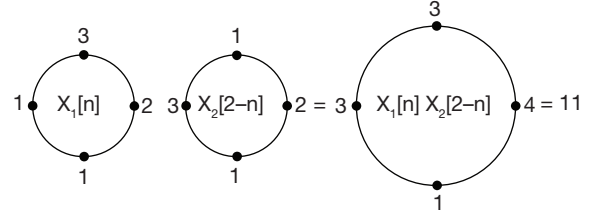
$$x_3(0) = x_1[n] x_2[-n] =$$



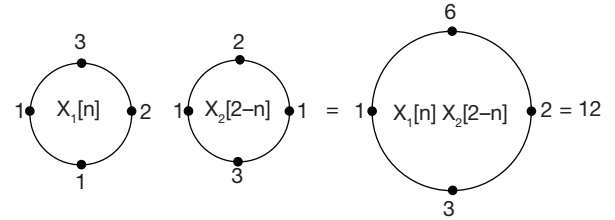
$$x_3(1) = x_1[n] x_2[1-n]$$



$$x_3(2) = x_1[n] x_2[2-n]$$



$$x_3(3) = x_1[n] x_2[3-n]$$



$$x_3[n] = [12, 14, 11, 12]$$

Choice (D)

11. $y[n] = x[n] + x[n-3] + x[n-5]$

By taking DTFT –

$$Y(e^{j\Omega}) = X(e^{j\Omega}) + e^{-3j\Omega} X(e^{j\Omega}) + e^{-5j\Omega} X(e^{j\Omega})$$

$$= (1 + e^{-3j\Omega} + e^{-5j\Omega}) X(e^{j\Omega})$$

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = (1 + e^{-3j\Omega} + e^{-5j\Omega})$$

$$= e^{-4j\Omega} (e^{4j\Omega} + e^{j\Omega} + e^{-j\Omega})$$

$$= e^{-4j\Omega} (e^{4j\Omega} + 2\cos\Omega)$$

$$= 1 + 2\cos\Omega e^{-4j\Omega}$$

Choice (D)

12. $X_{DFT}\{k\} = \{4, -j, 0, j\}$

here $N = 4$, so

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_{DFT}[k] e^{j\frac{2\pi nk}{4}}$$

$$x[n] = \frac{1}{4} \sum_{k=0}^3 X_{DFT}[k] e^{j\frac{2\pi nk}{4}}$$

$$\text{at } n = 0, x[0] = \frac{1}{4} \sum_{k=0}^3 X_{DFT}[k] e^{j0}$$

$$= \frac{1}{4} [4 - j + 0 + j] = 1$$

$$\text{at } n = 1, x[1] = \frac{1}{4} \sum_{k=0}^3 \left[4 - je^{\frac{j\pi}{2}} + 0 + je^{\frac{j3\pi}{2}} \right]$$

$$= \frac{1}{4} [4 + 1 + 0 + 1] = 1.5$$

$$\text{at } n = 2, x[2] = \frac{1}{4} \sum_{k=0}^3 X_{DFT}[k] e^{jk\pi}$$

$$= \frac{1}{4} [4 - je^{j\pi} + 0 + je^{j3\pi}]$$

$$= \frac{1}{4} [4 + j + 0 - j] = 1$$

$$\text{at } n = 3, x[3] = \frac{1}{4} \sum_{k=0}^3 X_{DFT}[k] e^{jk\frac{3\pi}{2}}$$

$$= \frac{1}{4} \left[4 - je^{\frac{j3\pi}{2}} + 0 + je^{\frac{j9\pi}{2}} \right] = 0.5$$

$$x[n] = \{1, 1.5, 1, 0.5\} \quad \text{Choice (B)}$$

13. As we know that by frequency shifting property –

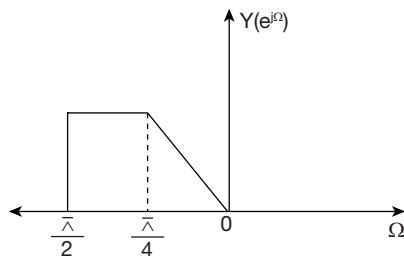
$$e^{j\Omega_0 n} x[n] \xleftrightarrow{DTFT} X[e^{j(\Omega - \Omega_0)}]$$

$$\text{So } e^{-jn\frac{\pi}{4}} x[n] = X\left(e^{j\left(\Omega + \frac{\pi}{4}\right)}\right)$$

So $Y(e^{j\Omega})$ will be the frequency advance version of

$$X(e^{j\Omega}) \text{ by } \frac{\pi}{4}$$

So, $Y(e^{j\Omega})$ will be



Choice (B)

14. (a) $x[n] = \left(\frac{1}{3}\right)^n u(n)$

It is a discrete signal and non-periodic signal. So discrete time fourier transform will be the appropriate Fourier Transform. (a) – (ii)

(b) $x(t) = 1 + \cos 3\pi t + \sin 5\pi t$

It is a continuous time signal and periodic also. So Fourier Series will be appropriate representation.

(b) – (i)

(c) $x(t) = e^{-2t} \cos 4\pi t u(t)$

It is continuous time signal and non-periodic also. So, Fourier Transform will be appropriate representation.

(c) – (iii)

(d) $x[n] = \sum_{m=-\infty}^{+\infty} \delta[n - 20m] - 2\delta[n - 3 - 20m]$

It is discrete time signal with period 20m. So, Discrete time fourier series will be appropriate representation. (d) – (iv) Choice (A)

15. If $P(j) = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$

$$\text{Then } p(t) = \frac{\sin \omega_c t}{\pi t}$$

$$H(\omega) = P(\omega - 3\omega_c) + P(\omega + 3\omega_c)$$

$$\text{So } h(t) = p(t) e^{j3\omega_c t} + p(t) e^{-j3\omega_c t}$$

$$= p(t) [e^{j3\omega_c t} + e^{-j3\omega_c t}]$$

$$= \frac{\sin \omega_c t}{\pi t} [e^{j3\omega_c t} + e^{-j3\omega_c t}] = \frac{\sin \omega_c t}{\pi t} (2 \cos 3\omega_c t)$$

Choice (B)

16. Group delay $\tau(\omega) = \frac{-d}{d\omega} \angle H(e^{j\omega})$

$$\text{So } \angle H(e^{j\omega}) = -\int \tau(\omega) d\omega$$

$$\angle H\left(e^{j\frac{\pi}{2}}\right) = \frac{2\omega}{\omega} = \frac{\pi}{\omega}$$

$$\angle H\left(e^{j\frac{\pi}{2}}\right) = -2 \times \frac{\pi}{2} = -\pi$$

$$\text{and } \left| H\left(e^{j\frac{\pi}{2}}\right) \right| = 2$$

$$\text{So } Y(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega})$$

$$X(e^{j\omega}) = \frac{Y(e^{j\omega})}{H(e^{j\omega})} = \frac{Y(e^{j\omega})}{|H(e^{j\omega})| e^{j\angle H(e^{j\omega})}}$$

$$= \frac{2\pi \left[\delta\left(\omega - \frac{\pi}{2}\right) + \delta\left(\omega + \frac{\pi}{2}\right) \right] e^{-j\frac{3\pi}{4}}}{2e^{-j\pi}}$$

$$= \pi \left[\delta\left(\omega - \frac{\pi}{2}\right) + \delta\left(\omega + \frac{\pi}{2}\right) \right] e^{+j\frac{\pi}{4}}$$

$$\text{So } x(t) = \cos\left(\frac{\pi n}{2} + \frac{\pi}{4}\right)$$

Choice (A)

17. Since $h[n] = 0$ for $n < 0$, So system is causal

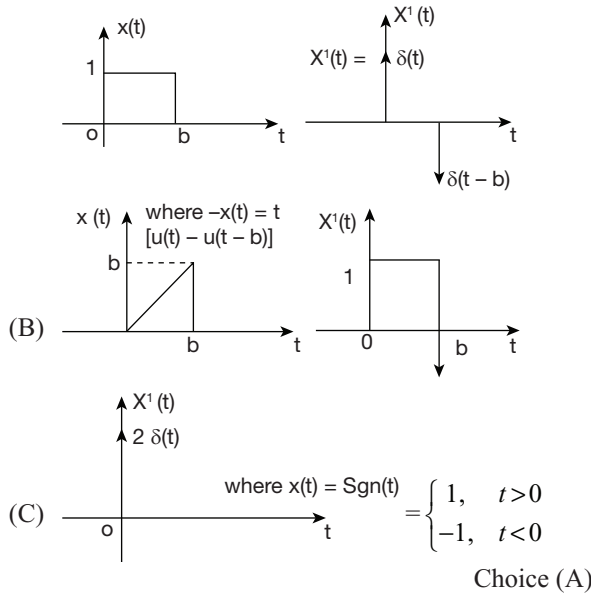
$$\sum_{k=-\infty}^{+\infty} h[k] = \sum_{k=-\infty}^{+\infty} |\beta^k u[n]|$$

$$= \sum_{k=0}^{\infty} |\beta|^k = \frac{1}{1-|\beta|} \quad |\beta| < 1$$

Therefore, the system is BIBO stable if $|\beta| < 1$ and unstable if $|\beta| \geq 1$. Choice (C)

18. Signal

(A) $x(t) = u(t) - u(t - b)$



19. (i) $y[n] = x^2[n]$
 for $x[n] = 2$ $y[n] = 4$
 for $x[n] = -2$ $y[n] = 4$
 for two distinct input output is same so, Non-invertible.
- (ii) $y[n] = \log\{x[n]\}$
 for $x[n] = 10^n$ and
 for $x[n] = 10^{-n}$ output
 is same $y[n] = \log\{10^{|n|}\} = n$
 So system is non-invertible.
- (iii) $y[n] = \sin\{x[n]\}$
 This system also produces same output for different inputs, So Non-invertible.
- (iv) $y[n] = x[n] + x[n-1]$
 This is invertible system and inverse is
 $x[n] = y[n] - y[n-1]$ Choice (D)
20. (A) $y[n] = 3^{n+2} x[n]$
 Output does not depend on future value of input so this is a causal system.
- (B) $y[n] = 3^n x[n+1]$
 at $n=0$, $y[0] = x[1]$
 output $y[0]$ depends on future value of input $\{x[1]\}$, So the system is not causal.
- (C) $y[n] = 3^n x[n]$
 for $n=1$, $y[1] = 3x[1]$
 output at $n=1$ depends on present value of input $\{x[2]\}$, So this is also a causal system.
 Choice (D)
21. **Causality:-**
 System is non causal because output depends on future value of input.
 at $n=1$, $y[1] = 5x[2]$
Linearity:-
 Let $x_1[2n] = 1$ then $y_1[n] = 0$
 $x_2[2n] = -1$ then $y_2[n] = -5$

for $x_3[n] = x_1[n] + x_2[n]$
 $y_3[n] \neq 0 \neq y_1[n] + y_2[n]$
 So system is not linear.

Inevitability:-

System is not invertible because for different input it has same outputs.

For example

$$y[n] = 0 \text{ for } x[n] = 1 \\ y[n] = 0 \text{ for } x[n] = 2$$

Choice (D)

$$22. H(z) = \frac{1}{1 - \frac{1}{3}Z^{-1}} + \frac{4}{1 - 2Z^{-1}}$$

$$H(z) = \frac{1}{1 - \frac{1}{3}Z^{-1}} + \frac{4}{1 - 2Z^{-1}}$$

The system has poles at $Z = \frac{1}{3}$ and $Z = 2$

Stability:

An LTI system is stable only if ROC of $H(z)$ contains unit circle so ROC: $\frac{1}{3} < |Z| < 2$

Causality:

For an LTI system to be causal the ROC must be exterior of a circle outside the outer most pole. Here outer most pole is $Z = 2$. So for a causal system ROC $|Z| > 2$.
 Choice (D)

23. For a causal system to be stable poles must lie in the negative half of S -plane.

- (A) $\frac{S-3}{(S+2)(S-1)}$ poles at $S = -2$ and at $S = 1$ poles lies in right half of S -plane. So it is unstable.
- (B) Poles at $S = 0$, $S = -3$ and $S = 1$ poles lie in right half of S -plane. So it is unstable.
- (C) Poles at $S = 0$, $S = -3$ and at $S = -2$ poles lies in left half of S -plane. So, it is stable. Choice (C)

$$24. x[k] = \sum_{n=0}^3 x[n] e^{\frac{-j2\pi kn}{4}} = \sum_{n=0}^2 x[n] e^{\frac{-j\pi kn}{2}} \\ = x(0)e^0 + x(1)e^{\frac{-j\pi k}{2}} + x(2)e^{-j\pi k} \\ = \frac{1}{3} + \frac{1}{3}e^{\frac{-j\pi k}{2}} + \frac{1}{3}e^{-j\pi k}$$

For 4-point DFT $x[k]$ has to be evaluated for $k = 0, 1, 2, 3$ –

$$\text{When } k = 0; X(0) = \frac{1}{3}$$

$$[1 + \cos 0 - j\sin 0 + \cos 0 - j\sin 0] \\ = \frac{1}{3} [1 + 1 + 1] = 1$$

When $k = 1$; $X(1)$

$$= \frac{1}{3} \left[1 + \cos \frac{\pi}{2} - j \sin \frac{\pi}{2} + \cos \pi - j \sin \pi \right]$$

$$= \frac{1}{3} [1 + 0 - j - 1 - j0] = \frac{-j}{3}$$

When $k = 2$; $X(2)$

$$= \frac{1}{3} [1 + \cos \pi - j \sin \pi + \cos 2\pi - j \sin 2\pi] = \frac{1}{2}$$

When $k = 3$; $X(3)$

$$= \frac{1}{3} \left[1 + \cos \frac{3\pi}{2} - j \sin \frac{3\pi}{2} - \right.$$

$$\left. j \sin \frac{3\pi}{2} + \cos 3\pi - j \sin 3\pi \right]$$

$$= \frac{1}{3} [1 + 0 + j - 1 - j0] = \frac{j}{3}$$

$$X[k] = \left\{ 1, \frac{-j}{3}, \frac{1}{3}, \frac{j}{3} \right\}$$

Choice (B)

$$25. y[n] = x[n] \cos 2\pi n$$

$$= x[n] \left[\frac{e^{-j2\pi n} + e^{+j2\pi n}}{2} \right]$$

$$= \frac{1}{2} x[n] e^{-j2\pi n} + \frac{1}{2} x[n] e^{+j2\pi n}$$

By using circular frequency shift property

$$x[n] e^{\frac{j2\pi n k_o}{N}} \xleftrightarrow{DFT} X(K - K_o)_N$$

here $K_o = 1$ $N = 1$

$$x[n] e^{j2\pi n} \xleftrightarrow{DFT} X(K - 1)_N$$

$$x[n] e^{-j2\pi n} \xleftrightarrow{DFT} X(K + 1)_N$$

DFT of $y[n]$

$$Y(k) = \frac{1}{2} [X(k - 1) + X(k + 1)]$$

Choice (B)